CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

November 2012

CL306 Transportation Engineering-II

Date: 05.11.2012, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Give a brief classification of Indian Railways. **[02]**

(b) Discuss the requirements of an ideal Permanent Way. **[05]**

OR

(b) Discuss the general measures/ways to increase the Capacity of a Railway Track. **[05]**

Q - 2 (a) Draw a cross section of Broad Gauge Railway Track in embankment (on Single Track). **[03]**

(b) Define Superelevation and mention the objectives of providing superelevation to Railway Tracks. Write the formulae of superelevation for B.G, M.G. & N.G. Also, discuss 'Negative Superelevation' with neat diagram. **[05]**

(c) Do as Directed:

(i) Define 'Cant Deficiency' & 'Grade Compensation'. **[03]**

(ii) Discuss the necessity of Points & Crossings in Railways. **[03]**

OR

Q - 2 (a) Mention different types of Track Gauges in Indian Railways along with their widths. **[03]**

(b) Discuss the necessity of Curves in Railways. Also write a short note on Types of curves. **[05]**

(c) Do as Directed:

(i) Derive the relationship of Superelevation (e) with Gauge (G), speed (V) and Radius of Curvature (R) with suitable assumptions and diagram. **[03]**

(ii) Write in brief the requirements of a good 'Crossing'? **[03]**

Q - 3 (a) Write a short note on 2 methods of laying sleepers on the points and crossings with neat diagram. **[02]**

(b) Define crossing. What are the component parts of a crossing? Discuss the requirements and characteristics of a good crossing. **[04]**

(c) Do as Directed:

(i) Discuss the objectives of Signaling in Railways. **[03]**

(ii) Enlist different types of light signals and explain any one. **[05]**

OR

Q - 3 (a) Classify the Station Yards. **[02]**

(b) Write a detailed note on Requirements of a Railway Station. **[04]**

(c) Do as Directed: [03]

(i) List out different types of Signals used in Indian Railways. [05]

(ii) Write a note on ware house and transit sheds with neat sketch.

SECTION – II

Q - 4 (a) Write functions of Airport Authority of India. [04]

(b) Define: **(Any Three)**

(i) Littoral drift (ii) Artificial harbor (iii) Erosion (iv) Silting. [03]

Q - 5 (a) Give the classification of flying activities. [04]

(b) Discuss in brief graving dry dock with neat sketch. [05]

(c) Enlist the criteria for the site selection for an air port. [05]

OR

Q - 5 (a) Discuss in brief minimum turning radius with neat sketch. [04]

(b) Define fenders. Enlist different types of fenders with neat sketch. [05]

(c) How to prepare the master plan for an airport. [05]

Q - 6 (a) How to decide the runway orientation. [03]

(b) Calculate the actual length of the runway from the following data: [07]

Airport elevation : R. L. 100

Airport reference temperature : 28° C

Basic runway length: 600 m

Highest point along the length : R. L. 98

Lowest point along the length : R. L. 95

(c) Determine the radius for a taxiway for a supersonic aircraft to negotiate the curve at a turning speed of 60 kmph. The wheel base is 30 m and the wheel tread is 7.2 m. the airport is of B type as per ICAO [04]

OR

Q - 6 (a) Define: Head wind, tail wind, cross wind [03]

(b) An airport site at sea level with standard atmospheric conditions, the runway lengths required for take off and landing are 2000 m and 2400 m respectively. The proposed airport situated at an altitude of 150 m. if the airport reference temparture is 20° C and if the effective runway gradient is 0.5 percent, calculate the length of runway to be provided. [07]

(c) What is meant by basic runway length: discuss any two cases to be considered. [04]

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